

Basic Cyber Physical System Attack and Defense Testbed Test Plan

1. Introduction

1.1 Document Identifier

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1.2 Scope

This Level Test Plan defines the scope, approach, resources, schedule, responsibilities, risks, and deliverables for testing the Basic Cyber Physical System Attack and Defense Testbed.

The system under test is a multi-vendor industrial control testbed built around PLCs, historian and HMI systems, network monitoring tools, and a physical sensor interception module. The test effort focuses on verifying that:

- Each PLC can be configured and operated correctly
- EtherNet/IP, PROFINET, and Modbus TCP communication works as intended
- The historian receives, displays, and records data correctly
- Attack scripts produce repeatable and observable effects from the HMI
- Physical sensor and relay-based manipulations are measurable and reversible
- Logging and capture tools record system behavior accurately

1.3 References

- IEEE Std 829-2008, IEEE Standard for Software and System Test Documentation 829-2008.pdf
- Project Plan: Basic Cyber Physical System Attack and Defense Testbed Plan.pdf

1.4 Test Objectives

The objectives of testing are to:

- Confirm the testbed is functional across all supported PLC platforms
- Validate the ability to run repeatable cyber and physical attack scenarios
- Confirm that monitoring, logging, and historian output are accurate
- Verify that changes in PLC, protocol, or sensor configuration can be observed and compared consistently
- Identify integration failures early and document anomalies clearly

2. Details for This Level of Test Plan

2.1 Test Items

The following items are in scope for testing:

- Siemens PLC
- Schneider Electric / Modicon PLC
- Allen-Bradley PLC
- EtherNet/IP communication
- PROFINET communication
- Modbus TCP communication
- HMI and historian systems
- Network switch configuration
- Sensor interception module
- ESP32 / Arduino Nano relay control path
- Logging and packet capture tools
- Attack scripts and test scripts

Excluded from this test effort:

- Non-project software unrelated to the testbed
- Long-term performance certification beyond basic validation
- Field deployment tests in real industrial environments

2.2 Test Approach

Testing will be performed in four levels:

1. Component testing
2. Component integration testing
3. System testing
4. Demonstration/acceptance testing

The approach is primarily black-box at the system level, with configuration and interface checks at the integration level. Test cases will include:

- Normal operation
- Negative tests
- Connection loss/recovery
- Protocol mismatch or misconfiguration
- Sensor interruption scenarios
- Repeatability checks for attack scripts

Test results will be compared against the expected PLC state, the predicted result after HMI runs, historian records, and packet capture evidence.

2.3 Test Environment

The test environment will include:

- EGSMTPC mini computer
- D-Link and TP-Link switches
- Siemens PLC
- Schneider Electric PLC
- Allen-Bradley PLC
- Sensor and switch hardware
- Arduino Nano / ESP32 relay module
- Wireshark, Scapy
- PLC programming tools:
 - Studio 5000 / RSLogix
 - Siemens TIA Portal / STEP 7
 - Schneider EcoStruxure / SoMachine Basic

The environment must support:

- Stable device addressing
- Isolated lab network
- Packet capture access
- Repeated reset to the known baseline state

2.4 Test Deliverables

The following deliverables will be produced:

- Test cases
- Test procedures
- Test execution log
- Anomaly reports
- Interim test status notes
- Final test report
- Evidence package including screenshots, packet captures, and log files

3. Test Management

3.1 Responsibilities

- Zach Lavoie:
 - Network and protocol connectivity tests
 - Historian communication tests
 - Packet capture validation
 - Cyber attack script testing with HMI

- Karla Daley:
 - PLC setup and configuration tests
 - Sensor and I/O validation
 - Hardware interception module tests
 - Physical response and reset tests
- Faculty Advisor:
 - Review and approval guidance
 - Technical validation support

3.2 Schedule

Milestone-based testing will follow the project schedule:

- Milestone 1: Initial setup and technical evaluation
 - Basic connectivity and “hello world” tests
 - Tool selection verification
 - Requirement and test planning validation
- Milestone 2: Feature implementation and testing
 - Internal attack vector testing
 - Sensor-to-PLC testing across all three PLCs
- Milestone 3: Final implementation and launch prep
 - External attack vector testing
 - ESP32 hardware attack testing

3.3 Risks and Contingencies

Key risks include:

- Multi-vendor PLC configuration complexity
- Protocol incompatibility
- HMI/historian integration delays
- Sensor interception module timing or wiring issues
- Repeatability issues across hardware configurations

Contingencies:

- Maintain rollback configurations for each PLC
- Use a known-good baseline for each test cycle
- Capture evidence at every test run
- Isolate failures to one subsystem at a time

3.4 Pass/Fail Criteria

A test passes if:

- The expected behavior matches the observed result

- PLC state changes are correct
- Network and historian logs match the event
- The test can be repeated with consistent results
- Any anomaly is recorded and understood

A test fails if:

- Communication does not establish correctly
- Physical or cyber manipulation causes unplanned behavior without explanation
- Logs do not match observed behavior
- The system cannot be restored to baseline

4. General

4.1 Metrics

The following metrics will be tracked:

- Number of successful test cases
- Number of failed test cases
- Number of anomalies discovered
- Time to recover to baseline
- Repeatability rate of attack scripts
- PLC-to-Historian logging consistency

4.2 Change Control

All changes to this document shall be recorded by version number, date, author, and reason for change. Updated versions will be stored in the project repository.

4.3 Glossary

- PLC: Programmable Logic Controller
- HMI: Human-Machine Interface computer
- Historian: Computer that logs & manages the PLCs
- I/O: Input/Output
- Attack script: Reusable test script used to simulate a cyber event
- Baseline: Known-good system state before test execution